

TASK 4: COMPUTER VISION OBJECT DETECTOR

Progree Remote Artificial Intelligence Internship | Technical Whitepaper

1. Executive Summary & Objective

Program: Progree AI Internship | Task: 4 (Computer Vision)

Domain: Object Detection and Segmentation | Source: Deterministic Synthetic Video

This project implements a reproducible OpenCV pipeline that extracts target contours from video frames, classifies synthetic shapes, tracks them across time, overlays dynamic counts, and exports frame-level inference metrics.

Key outcomes

90 frames processed	162.3 mean FPS	6.45 mean ms/frame	0.88 mean confidence
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Synthetic video -> adaptive threshold -> contours -> color classes -> centroid tracking -> annotated video + CSV

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2. Pipeline Architecture & Methodology

Classical CV processing chain

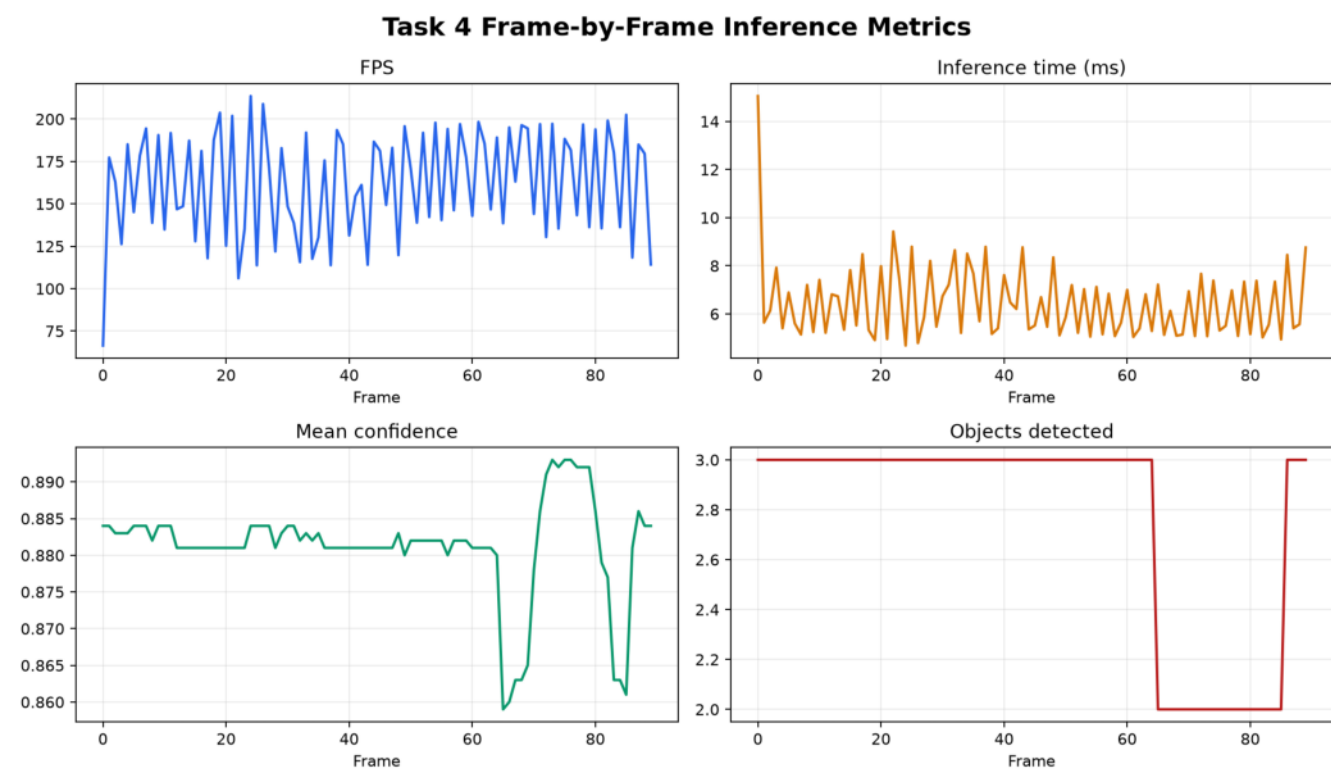
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|----|--------------|---|
| 01 | Video input | OpenCV reads a deterministic 640x360 MP4. |
| 02 | Filtering | Gaussian blur suppresses high-frequency noise. |
| 03 | Segmentation | Adaptive Gaussian threshold builds local pixel-bound masks. |
| 04 | Contours | Morphology and external contours produce regions. |
| 05 | Tracking | Nearest-centroid association preserves IDs. |
| 06 | Output | Overlay, MP4, sample PNGs, and CSV metrics. |

The detector boundary is intentionally isolated from tracking and reporting. A YOLOv8n adapter can replace contour extraction for natural classes without changing the rest of the pipeline.

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3. Frame-by-Frame Performance Metrics



Mean FPS: 162.3 | Mean inference time: 6.45 ms | Mean confidence: 0.881 | Mean objects/frame: 2.77

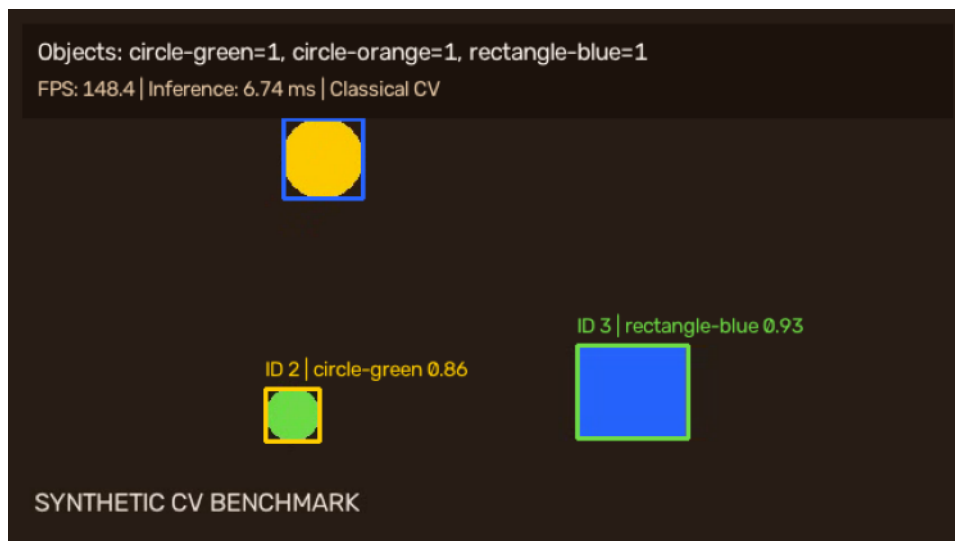
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4. Annotated Frame Evidence



frame_000.png



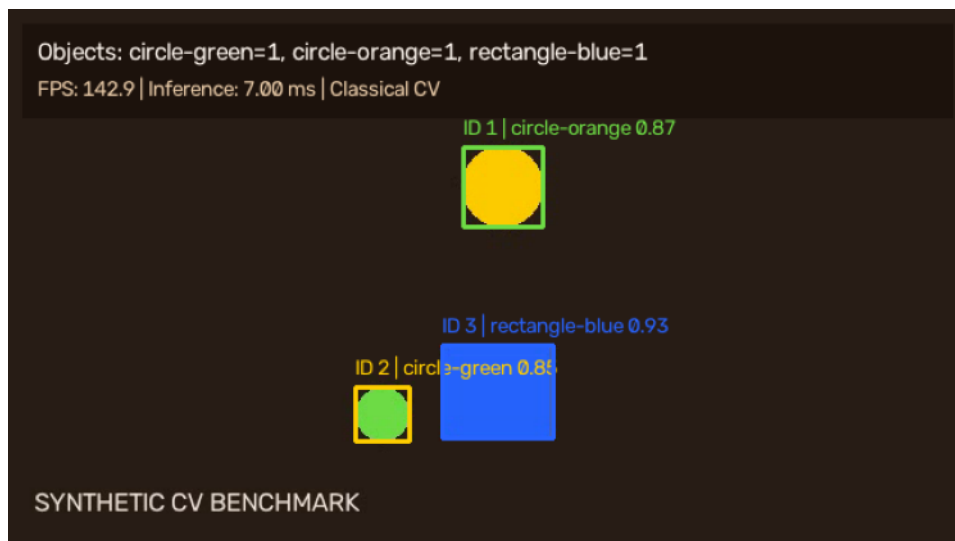
frame_030.png

Dynamic overlay includes class counts, track IDs, confidence, FPS, inference time, and the active classical CV detector label.

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4. Annotated Frame Evidence



frame_060.png



frame_089.png

Dynamic overlay includes class counts, track IDs, confidence, FPS, inference time, and the active classical CV detector label.

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5. Accuracy, Performance & Reproduction

Trade-off discussion

Classical CV is fast, explainable, and lightweight, but its color and contour rules are domain-specific. Shadows, occlusion, texture, and similar backgrounds can reduce recall or merge regions.

A YOLOv8n-style model would improve semantic generalization for natural classes, at the cost of model weights, startup time, and higher latency. A hybrid system can combine learned detections with classical masks.

Reproduce with: `py -3.14 main.py`, then `py -3.14 -m pytest tests -q`. Outputs include `annotated_output.mp4`, four sample PNGs, `metrics_dashboard.png`, and `frame_metrics.csv`.